JavaScript RegExp

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Regular Expression Syntax

A regular expression is a sequence of characters that forms a search pattern. It is written between slashes `/pattern/`:

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Regular Expression Syntax</h4>
<p id="demo"></p>

<script>
const text = "The rain in Spain falls mainly in the plain";

// Search for "ain" using a regular expression
const pattern = /ain/;

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "Pattern: /ain/<br>" +
  "match() result: " + text.match(pattern);
</script>

</body>
</html>
```

Example 1

Result [View Example]

Using String Methods

Regular expressions are commonly used with string methods: `match()`, `replace()`, and `search()`:

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>Using String Methods</h4>
<p id="demo"></p>

<script>
const text = "Visit W3Schools";

// match() - returns matches
const matchResult = text.match(/W3Schools/);

// replace() - replaces matches
const replaceResult = text.replace(/Visit/, "Welcome to");

// search() - returns position
const searchResult = text.search(/W3Schools/);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/W3Schools/): " + matchResult + "<br>" +
  "replace(/Visit/, 'Welcome to'): " + replaceResult + "<br>" +
  "search(/W3Schools/): " + searchResult + " (position)";
</script>

</body>
</html>
```

Example 2

Result [View Example]

JavaScript Regex Flags

Flags modify how the pattern is searched:

Flag	Description
`/g`	Global - matches all occurrences
`/i`	Insensitive - case insensitive
`/m`	Multiline - ^ and $ match line breaks

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JavaScript Regex Flags</h4>
<p id="demo"></p>

<script>
const text = "The rain in Spain";

// Without /g - only first match
const noG = text.match(/ain/);

// With /g - all matches
const withG = text.match(/ain/g);

// With /i - case insensitive
const withI = text.match(/THE/i);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/ain/): " + noG + " (first only)<br>" +
  "match(/ain/g): " + withG + " (all occurrences)<br>" +
  "match(/THE/i): " + withI + " (case insensitive)";
</script>

</body>
</html>
```

Example 3

Result [View Example]

RegExp Metacharacters

Metacharacters have special meanings in regular expressions:

Metacharacter	Description
`\d`	Find a digit
`\s`	Find a whitespace
`\w`	Find a word character
`.`	Find a single character
`\b`	Find at word boundary

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript RegExp</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>RegExp Metacharacters</h4>
<p id="demo"></p>

<script>
const text = "Give 100% and 25% effort";

// \d finds digits
const digits = text.match(/\d/g);

// \w finds word characters
const words = text.match(/\w+/g);

document.getElementById("demo").innerHTML =
  "Text: " + text + "<br><br>" +
  "match(/\\d/g) digits: " + digits + "<br>" +
  "match(/\\w+/g) words: " + words;
</script>

</body>
</html>
```

Example 4

Result [View Example]

Document

Document in project

You can Download PDF file.

Reference

- W3Schools JavaScript RegExp